

Reproducibility Supplement

Training and Calibration Costs of Innovation-Normalized Prediction for Autoregressive Episodes

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1 Scope and provenance

This supplement documents the newly generated numerical study accompanying the revised manuscript. The earlier 27 September PDFs are preserved under [paper/archive/2026-09-27/](#). The original comparator programs and raw outputs were unavailable in the retrieved repository. Accordingly, this revision replaces their empirical claims with fresh, fully specified calculations; it does not claim to reproduce the earlier 4,800 fits.

The internal protocol `reproducibility/PROTOCOL.md` was written before this revision’s confirmation comparison. It is not an externally registered protocol. A two-fit-per-cell implementation pilot used seed 2026092802; none of those pilot outcomes enter the reported results. The confirmation study uses seed 2026092812. Source code, saved outcomes and the table generators are included in the repository. All findings are synthetic; no sensing deployment or real-data result is claimed.

2 Design and estimands

Each independent fitted repetition uses 500 training episodes, 499 calibration episodes and 1,000 evaluation episodes. The target miscoverage is .1, giving rank 450 and rank coverage .9. Histories have length 4 or 12. Four mechanisms and two history lengths, with 100 repetitions in each cell, give 800 independent fitted datasets. All eight procedures share the datasets within a cell. Separate random streams determine training, calibration, evaluation and model fitting.

The common marginal scale is $\sigma = \sqrt{3/G}$, $G \sim \text{Gamma}(4, 1)$, independent across episodes and independent of standardized shape. Thus $\mathbb{E}\sigma^2 = 1$ and $\mathbb{E}\sigma^4 = 1.5$. The standardized mechanisms are:

1. Gaussian stationary AR(1), coefficient .8 and innovation variance .36.
2. AR(1), coefficient .8, standardized Student- t_5 innovations multiplied by .6; discard 512 initialization steps. The resulting finite burn-in is an approximation to stationarity.
3. Gaussian AR(2), coefficients (.5,.3), innovation variance .4457142857142857; discard 512 initialization steps.
4. Gaussian AR(1), independent episode coefficient chosen equally from $\{-.8, .8\}$, with stationary initialization conditional on that coefficient.

Training, calibration and evaluation share the same standardized mechanism in each structural study. A paired amplitude arm doubles every coordinate in each evaluation episode. An additional Gaussian negative control doubles only the future response, leaving its history unchanged. This last arm does not satisfy the scale-transfer premise.

The primary estimands are the mean, over fitted datasets, of test coverage and mean interval length. The reported Monte Carlo standard errors use the 100 fitted repetitions as independent units. The 1,000 observations within a fitted test set are not treated as independent fitted models. For paired length means $\bar{X}, \bar{Y}$, let $r = \bar{X}/\bar{Y}$. The ratio standard error is

$$\widehat{\text{se}}(r) = \frac{s(X - rY)}{\sqrt{R} \bar{Y}}, \quad R = 100.$$

Intervals use $r \pm 1.96\widehat{\text{se}}(r)$ and are descriptive Monte Carlo intervals, not simultaneous inference across all comparisons. Width is zero for an empty set, and empty-set frequency is retained. No unfavorable cell is removed. Training times in the raw file are host-specific: AR ablations share the coefficient fit, and the Student timing records that fit only. These times are not used to rank methods.

3 Comparator specification and verification

IN-ARCP clips the pooled estimate to $[-.98, .98]$ and uses the history innovation scale. The raw-RMS ablation changes only that scale to $\sqrt{m^{-1} \sum_j H_j^2}$. Global AR calibration uses the same center and an unnormalized absolute residual. The Student reference uses the fitted AR center and innovation scale with $t_{m,.95}$; it is an exact pivot only when the coefficient equals the true Gaussian parameter.

For invariant learned procedures, define $A(H) = \sqrt{m^{-1} \sum_j H_j^2}$, train on $(H/A(H), Y/A(H))$, and multiply predicted endpoints by $A(H)$ on the original scale. Native Ad-EffO instead receives the original coordinates. Both use an intercept and all history coordinates. The independently written smoothed absolute-error-quantile gradient uses 1,000 updates, steps $t^{-.6}$, smoothing .1, and compact polynomial weights proportional to $[1 - (d/.1)^2]^2$ for $|d| < .1$. Its residual quantile forest has 100 trees, maximum depth 5 and quantile .9. This follows the published four-step construction with stated settings; it does not assert optimal tuning or exact reproduction of the authors' benchmark.

The linear update was checked against upstream `utils.py`, Git blob `58d4bf1c69002d0eab982fa5b25beb574434bf71`, at commit `025118446ab9636837e91602efe4fa5e138f2373`. Initial exploratory examples matched after 1,000 updates, while intermediate trajectories sometimes differed. A further fully specified three-seed check, now distributed as `check_comparator.py`, gives initial-update agreement within 2.3×10^{-16} but a final coefficient difference of .229 on one dataset. Thus equality of optimizer trajectories is not claimed. Algebraically identical polynomial weights and different floating-point reduction orders can alter this nonsmooth optimization path. The original exploratory checks and the subsequent check are both retained. The latter records its complete data-generation design and verifies the upstream Git blob before executing the named functions.

This finding prompted a post-validation sensitivity analysis on the same 200 Gaussian fitted datasets used in the primary study. It evaluates 1,000 and 2,000 updates with both the simplified polynomial arithmetic used in the main experiment and the unscaled polynomial arithmetic in the upstream expression. Both are independently vectorized; neither is asserted to reproduce the exact upstream loop's trajectory. All 16 aggregated combinations are reported in the final section. This follow-up does not retroactively change the original confirmation protocol or select a replacement winning comparator. The upstream linear notebook also uses different illustrative settings, including a .8 residual quantile. Upstream source is not redistributed.

Invariant CQR uses a quantile forest with 100 trees, depth 5, quantiles .05 and .95, followed by corrected additive calibration of $\max(\hat{q}_{.05}(H) - Y, Y - \hat{q}_{.95}(H))$. The invariant nonlinear RF procedure uses a center forest with 100 trees, depth 8 and minimum leaf size 10; a residual-scale forest has 100 trees, depth 5 and minimum leaf size 10. Its absolute normalized residual is calibrated multiplicatively. No hyperparameter is selected on the evaluation episodes. These comparisons address specified implementations and information sets, rather than the best achievable performance of every method family.

4 Numerical mathematics and planning

Whitening, determinant and directional-variance identities are checked over $m \in \{2, 3, 4, 12, 25\}$ and $\rho \in \{-.95, -.4, 0, .4, .95\}$. Known-coefficient calibration means are checked over $m \in \{2, 4, 12\}$ and calibration sizes

$$n \in \{9, 19, 49, 50, 99, 100, 199, 200, 499, 1999\}.$$

The Jacobi rules have 128 and 512 nodes and are compared with independently implemented adaptive beta-density integration. The public utility defaults to adaptive tail integration because inverse-CDF quadrature converges slowly at extreme ranks. No fixed node count is claimed universally accurate.

Four complete fitted-mean calculations use $m = 2$. The coarse integration uses 64 coefficient bins, 256 midpoint angles and 32 beta nodes; the refinement uses 128 bins, 512 angles and 64 beta nodes. Both clipped coefficient atoms are retained. Each mean is checked against 30,000 independent fits and 32 test observations per fit, with episodes generated using a covariance Cholesky factor. Seed: 2026092801. The maximum discrepancy is 1.42 Monte Carlo standard errors. Grid refinement is a numerical diagnostic, not a certified error bound.

The local population curvature is checked at five coefficients with 4,096 circle directions and coefficient differences 10^{-4} . Coefficient MSE is evaluated using 10,000 repetitions at every (m, κ, N) in $\{4, 12\} \times \{1, 1.5, 3\} \times \{20, 100, 500\}$. The heavy-scale discrepancy at $\kappa = 3$ remains in the results.

The allocation experiment uses a total of 300 episodes, $m = 4$, $\rho = .8$ and $\kappa \in \{1, 1.5\}$. Planning parameters are fixed from the synthetic design. There are 1,000 paired fitted repetitions per candidate, with 500 evaluation episodes, seed stream based on 2026092803. Candidate training sizes are 21, 51, 71, 101, 150, 201, 251, plus the planning recommendation 61 when $\kappa = 1$. A shared 300-episode pool is partitioned at each candidate boundary. The observed test outcomes do not choose the reported planning candidate. The full candidate table is supplied below; it does not establish a unique finite-sample optimum.

5 Software and execution

The saved comparison was run with Python 3.12.14, NumPy 2.3.5, SciPy 1.17.0, scikit-learn 1.8.0 and sklearn-quantile 0.1.1. Figure generation used Matplotlib 3.10.8. The four-worker comparison took approximately 848 seconds on the execution host; runtime is not a hardware-independent claim. Exact floating-point and model-fitting outcomes may depend on platform and package builds.

From the repository root, install `reproducibility/requirements.txt` and the package, then run the following commands. Existing saved outputs support rebuilding figures and PDFs without rerunning the simulations.

```
python -m pip install -r reproducibility/requirements.txt
python -m pip install -e .
python -m unittest discover -s tests -v
python reproducibility/validate_mathematics.py
python reproducibility/validate_expansion.py
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 \
  python reproducibility/run_comparisons.py --repetitions 100 --workers 4
python reproducibility/run_planning.py
python reproducibility/build_artifacts.py
python reproducibility/check_optimizer_sensitivity.py
python reproducibility/build_optimizer_summary.py
python reproducibility/build_supplement.py
cd paper
latexmk -pdf -interaction=nonstopmode -halt-on-error manuscript.tex
latexmk -pdf -interaction=nonstopmode -halt-on-error supplement.tex
```

The main manuscript uses the IEEE Access document class and IEEE numbered references. This supplement uses a separate, single-column layout for the full result tables. Template dependencies and their provenance are documented in `paper/TEMPLATE.md`. SHA256SUMS records the distributed evidence files. Numerical assertions and code tests support computational correctness; they do not establish priority, application utility or publication acceptance.

6 Complete comparison results

“same” is the unchanged evaluation amplitude, “scale_x2” doubles the whole episode, and “future_x2” doubles only its response. Parentheses contain Monte Carlo standard errors. The ratio numerator is always IN-ARCP. An empty fraction rounded to zero is not proof that emptiness is impossible. The machine-readable files preserve all digits and all per-fit measurements.

Method	Coverage (MCSE)	Length (MCSE)	IN/method [95% MC]	Empty
gaussian, $m = 4$, same				
IN-ARCP	0.8985 (0.0018)	2.318 (0.014)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.8998 (0.0019)	2.799 (0.020)	0.828 [0.819, 0.837]	0.000
AR+global	0.8978 (0.0018)	1.933 (0.009)	1.199 [1.184, 1.213]	0.000
Student	0.8988 (0.0010)	2.309 (0.004)	1.004 [0.992, 1.016]	0.000
Ad-EffOrt native	0.8975 (0.0020)	2.017 (0.010)	1.149 [1.136, 1.162]	0.000
Ad-EffOrt invariant	0.9002 (0.0017)	2.774 (0.018)	0.835 [0.825, 0.845]	0.000
CQR invariant	0.8977 (0.0019)	2.825 (0.021)	0.820 [0.811, 0.830]	0.000
RF invariant	0.8990 (0.0018)	2.649 (0.018)	0.875 [0.865, 0.885]	0.000
gaussian, $m = 4$, scale_x2				
IN-ARCP	0.8985 (0.0018)	4.635 (0.028)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.8998 (0.0019)	5.599 (0.041)	0.828 [0.819, 0.837]	0.000
AR+global	0.6203 (0.0026)	1.933 (0.009)	2.398 [2.369, 2.427]	0.000
Student	0.8988 (0.0010)	4.619 (0.008)	1.004 [0.992, 1.016]	0.000
Ad-EffOrt native	0.7016 (0.0035)	2.490 (0.018)	1.862 [1.832, 1.892]	0.000
Ad-EffOrt invariant	0.9002 (0.0017)	5.548 (0.036)	0.835 [0.825, 0.845]	0.000
CQR invariant	0.8977 (0.0019)	5.650 (0.041)	0.820 [0.811, 0.830]	0.000
RF invariant	0.8990 (0.0018)	5.298 (0.036)	0.875 [0.865, 0.885]	0.000

Method	Coverage (MCSE)	Length (MCSE)	IN/method [95% MC]	Empty
gaussian, $m = 4$, future_x2				
IN-ARCP	0.5708 (0.0031)	2.318 (0.014)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.6359 (0.0036)	2.799 (0.020)	0.828 [0.819, 0.837]	0.000
AR+global	0.5377 (0.0025)	1.933 (0.009)	1.199 [1.184, 1.213]	0.000
Student	0.5700 (0.0016)	2.309 (0.004)	1.004 [0.992, 1.016]	0.000
Ad-EffOort native	0.5424 (0.0029)	2.017 (0.010)	1.149 [1.136, 1.162]	0.000
Ad-EffOort invariant	0.6309 (0.0035)	2.774 (0.018)	0.835 [0.825, 0.845]	0.000
CQR invariant	0.6292 (0.0036)	2.825 (0.021)	0.820 [0.811, 0.830]	0.000
RF invariant	0.6148 (0.0034)	2.649 (0.018)	0.875 [0.865, 0.885]	0.000
gaussian, $m = 12$, same				
IN-ARCP	0.8991 (0.0015)	2.008 (0.008)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.8986 (0.0014)	2.253 (0.010)	0.891 [0.884, 0.898]	0.000
AR+global	0.8972 (0.0017)	1.923 (0.009)	1.044 [1.036, 1.052]	0.000
Student	0.9006 (0.0010)	2.012 (0.002)	0.998 [0.991, 1.005]	0.000
Ad-EffOort native	0.8975 (0.0018)	2.044 (0.009)	0.983 [0.974, 0.991]	0.000
Ad-EffOort invariant	0.9001 (0.0015)	2.387 (0.012)	0.841 [0.832, 0.850]	0.000
CQR invariant	0.8985 (0.0015)	2.358 (0.010)	0.852 [0.845, 0.859]	0.000
RF invariant	0.9001 (0.0016)	2.318 (0.011)	0.866 [0.858, 0.874]	0.000
gaussian, $m = 12$, scale_x2				
IN-ARCP	0.8991 (0.0015)	4.016 (0.016)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.8986 (0.0014)	4.507 (0.020)	0.891 [0.884, 0.898]	0.000
AR+global	0.6175 (0.0026)	1.923 (0.009)	2.088 [2.071, 2.105]	0.000
Student	0.9006 (0.0010)	4.024 (0.004)	0.998 [0.991, 1.005]	0.000
Ad-EffOort native	0.7470 (0.0029)	2.868 (0.021)	1.400 [1.379, 1.422]	0.000
Ad-EffOort invariant	0.9001 (0.0015)	4.774 (0.024)	0.841 [0.832, 0.850]	0.000
CQR invariant	0.8985 (0.0015)	4.715 (0.021)	0.852 [0.845, 0.859]	0.000
RF invariant	0.9001 (0.0016)	4.635 (0.021)	0.866 [0.858, 0.874]	0.000
gaussian, $m = 12$, future_x2				
IN-ARCP	0.5223 (0.0022)	2.008 (0.008)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.5597 (0.0022)	2.253 (0.010)	0.891 [0.884, 0.898]	0.000
AR+global	0.5350 (0.0024)	1.923 (0.009)	1.044 [1.036, 1.052]	0.000
Student	0.5235 (0.0015)	2.012 (0.002)	0.998 [0.991, 1.005]	0.000
Ad-EffOort native	0.5449 (0.0025)	2.044 (0.009)	0.983 [0.974, 0.991]	0.000
Ad-EffOort invariant	0.5788 (0.0028)	2.387 (0.012)	0.841 [0.832, 0.850]	0.000
CQR invariant	0.5725 (0.0025)	2.358 (0.010)	0.852 [0.845, 0.859]	0.000
RF invariant	0.5702 (0.0024)	2.318 (0.011)	0.866 [0.858, 0.874]	0.000
heavy, $m = 4$, same				
IN-ARCP	0.8991 (0.0017)	2.358 (0.016)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9022 (0.0016)	2.786 (0.020)	0.847 [0.836, 0.857]	0.000
AR+global	0.8988 (0.0018)	1.846 (0.011)	1.278 [1.258, 1.297]	0.000
Student	0.8882 (0.0010)	2.217 (0.004)	1.064 [1.050, 1.078]	0.000
Ad-EffOort native	0.8983 (0.0016)	1.959 (0.012)	1.204 [1.187, 1.221]	0.000
Ad-EffOort invariant	0.8996 (0.0018)	2.835 (0.025)	0.832 [0.820, 0.843]	0.000
CQR invariant	0.9004 (0.0016)	2.905 (0.020)	0.812 [0.802, 0.822]	0.000
RF invariant	0.8995 (0.0016)	2.713 (0.022)	0.869 [0.858, 0.880]	0.000
heavy, $m = 4$, scale_x2				
IN-ARCP	0.8991 (0.0017)	4.717 (0.033)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9022 (0.0016)	5.572 (0.040)	0.847 [0.836, 0.857]	0.000
AR+global	0.6642 (0.0031)	1.846 (0.011)	2.555 [2.517, 2.594]	0.000
Student	0.8882 (0.0010)	4.435 (0.009)	1.064 [1.050, 1.078]	0.000
Ad-EffOort native	0.7303 (0.0029)	2.438 (0.021)	1.935 [1.900, 1.970]	0.000
Ad-EffOort invariant	0.8996 (0.0018)	5.670 (0.051)	0.832 [0.820, 0.843]	0.000
CQR invariant	0.9004 (0.0016)	5.810 (0.041)	0.812 [0.802, 0.822]	0.000
RF invariant	0.8995 (0.0016)	5.427 (0.044)	0.869 [0.858, 0.880]	0.000

Method	Coverage (MCSE)	Length (MCSE)	IN/method [95% MC]	Empty
heavy, $m = 12$, same				
IN-ARCP	0.9029 (0.0017)	2.017 (0.012)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9018 (0.0016)	2.235 (0.014)	0.903 [0.895, 0.910]	0.000
AR+global	0.8988 (0.0018)	1.829 (0.011)	1.103 [1.090, 1.116]	0.000
Student	0.8978 (0.0010)	1.962 (0.003)	1.028 [1.016, 1.040]	0.000
Ad-EffOrr native	0.9001 (0.0016)	1.996 (0.012)	1.011 [0.999, 1.023]	0.000
Ad-EffOrr invariant	0.9006 (0.0015)	2.415 (0.014)	0.835 [0.827, 0.844]	0.000
CQR invariant	0.9020 (0.0016)	2.403 (0.015)	0.840 [0.832, 0.848]	0.000
RF invariant	0.9013 (0.0016)	2.312 (0.015)	0.873 [0.864, 0.881]	0.000
heavy, $m = 12$, scale_x2				
IN-ARCP	0.9029 (0.0017)	4.035 (0.025)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9018 (0.0016)	4.470 (0.028)	0.903 [0.895, 0.910]	0.000
AR+global	0.6628 (0.0032)	1.829 (0.011)	2.206 [2.180, 2.232]	0.000
Student	0.8978 (0.0010)	3.925 (0.006)	1.028 [1.016, 1.040]	0.000
Ad-EffOrr native	0.7753 (0.0028)	2.853 (0.028)	1.414 [1.387, 1.442]	0.000
Ad-EffOrr invariant	0.9006 (0.0015)	4.830 (0.028)	0.835 [0.827, 0.844]	0.000
CQR invariant	0.9020 (0.0016)	4.805 (0.030)	0.840 [0.832, 0.848]	0.000
RF invariant	0.9013 (0.0016)	4.623 (0.029)	0.873 [0.864, 0.881]	0.000
ar2, $m = 4$, same				
IN-ARCP	0.8983 (0.0017)	2.666 (0.016)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.8980 (0.0017)	3.076 (0.021)	0.866 [0.857, 0.876]	0.000
AR+global	0.8996 (0.0018)	2.270 (0.011)	1.174 [1.160, 1.188]	0.000
Student	0.9014 (0.0009)	2.684 (0.005)	0.993 [0.983, 1.004]	0.000
Ad-EffOrr native	0.8991 (0.0015)	2.239 (0.009)	1.190 [1.177, 1.204]	0.000
Ad-EffOrr invariant	0.8989 (0.0017)	2.989 (0.020)	0.892 [0.882, 0.901]	0.000
CQR invariant	0.9007 (0.0018)	3.027 (0.019)	0.881 [0.872, 0.890]	0.000
RF invariant	0.8994 (0.0017)	2.877 (0.018)	0.927 [0.918, 0.935]	0.000
ar2, $m = 4$, scale_x2				
IN-ARCP	0.8983 (0.0017)	5.331 (0.032)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.8980 (0.0017)	6.153 (0.042)	0.866 [0.857, 0.876]	0.000
AR+global	0.6219 (0.0027)	2.270 (0.011)	2.348 [2.321, 2.376]	0.000
Student	0.9014 (0.0009)	5.368 (0.011)	0.993 [0.983, 1.004]	0.000
Ad-EffOrr native	0.7096 (0.0029)	2.818 (0.020)	1.892 [1.861, 1.923]	0.000
Ad-EffOrr invariant	0.8989 (0.0017)	5.979 (0.040)	0.892 [0.882, 0.901]	0.000
CQR invariant	0.9007 (0.0018)	6.053 (0.038)	0.881 [0.872, 0.890]	0.000
RF invariant	0.8994 (0.0017)	5.754 (0.037)	0.927 [0.918, 0.935]	0.000
ar2, $m = 12$, same				
IN-ARCP	0.9004 (0.0016)	2.364 (0.010)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9019 (0.0016)	2.632 (0.013)	0.898 [0.891, 0.905]	0.000
AR+global	0.8980 (0.0015)	2.260 (0.009)	1.046 [1.037, 1.054]	0.000
Student	0.8981 (0.0010)	2.340 (0.003)	1.010 [1.002, 1.018]	0.000
Ad-EffOrr native	0.8990 (0.0014)	2.288 (0.010)	1.033 [1.024, 1.042]	0.000
Ad-EffOrr invariant	0.9029 (0.0018)	2.658 (0.016)	0.889 [0.880, 0.898]	0.000
CQR invariant	0.9001 (0.0017)	2.601 (0.014)	0.909 [0.900, 0.917]	0.000
RF invariant	0.9004 (0.0018)	2.536 (0.016)	0.932 [0.923, 0.941]	0.000
ar2, $m = 12$, scale_x2				
IN-ARCP	0.9004 (0.0016)	4.727 (0.021)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9019 (0.0016)	5.263 (0.027)	0.898 [0.891, 0.905]	0.000
AR+global	0.6209 (0.0025)	2.260 (0.009)	2.091 [2.074, 2.109]	0.000
Student	0.8981 (0.0010)	4.681 (0.005)	1.010 [1.002, 1.018]	0.000
Ad-EffOrr native	0.7504 (0.0027)	3.207 (0.024)	1.474 [1.452, 1.496]	0.000
Ad-EffOrr invariant	0.9029 (0.0018)	5.316 (0.031)	0.889 [0.880, 0.898]	0.000
CQR invariant	0.9001 (0.0017)	5.202 (0.029)	0.909 [0.900, 0.917]	0.000
RF invariant	0.9004 (0.0018)	5.071 (0.033)	0.932 [0.923, 0.941]	0.000

Method	Coverage (MCSE)	Length (MCSE)	IN/method [95% MC]	Empty
variable, $m = 4$, same				
IN-ARCP	0.9008 (0.0016)	3.539 (0.019)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9010 (0.0016)	3.546 (0.019)	0.998 [0.996, 1.000]	0.000
AR+global	0.9004 (0.0015)	3.250 (0.016)	1.089 [1.074, 1.104]	0.000
Student	0.9071 (0.0009)	3.619 (0.007)	0.978 [0.968, 0.987]	0.000
Ad-EffOrr native	0.8976 (0.0018)	2.518 (0.011)	1.406 [1.389, 1.422]	0.000
Ad-EffOrr invariant	0.9010 (0.0016)	3.352 (0.021)	1.056 [1.046, 1.066]	0.000
CQR invariant	0.9011 (0.0017)	3.230 (0.022)	1.096 [1.085, 1.107]	0.000
RF invariant	0.8999 (0.0015)	3.028 (0.020)	1.169 [1.156, 1.182]	0.000
variable, $m = 4$, scale_x2				
IN-ARCP	0.9008 (0.0016)	7.078 (0.038)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9010 (0.0016)	7.092 (0.038)	0.998 [0.996, 1.000]	0.000
AR+global	0.6237 (0.0026)	3.250 (0.016)	2.178 [2.148, 2.207]	0.000
Student	0.9071 (0.0009)	7.238 (0.015)	0.978 [0.968, 0.987]	0.000
Ad-EffOrr native	0.7108 (0.0028)	3.248 (0.023)	2.179 [2.143, 2.215]	0.000
Ad-EffOrr invariant	0.9010 (0.0016)	6.703 (0.042)	1.056 [1.046, 1.066]	0.000
CQR invariant	0.9011 (0.0017)	6.459 (0.045)	1.096 [1.085, 1.107]	0.000
RF invariant	0.8999 (0.0015)	6.056 (0.040)	1.169 [1.156, 1.182]	0.000
variable, $m = 12$, same				
IN-ARCP	0.9020 (0.0017)	3.346 (0.015)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9018 (0.0017)	3.349 (0.015)	0.999 [0.996, 1.001]	0.000
AR+global	0.8985 (0.0015)	3.217 (0.016)	1.040 [1.029, 1.050]	0.000
Student	0.8892 (0.0010)	3.211 (0.005)	1.042 [1.033, 1.050]	0.000
Ad-EffOrr native	0.8993 (0.0017)	2.596 (0.013)	1.289 [1.275, 1.302]	0.000
Ad-EffOrr invariant	0.9016 (0.0016)	3.000 (0.015)	1.115 [1.105, 1.126]	0.000
CQR invariant	0.9007 (0.0016)	2.694 (0.014)	1.242 [1.229, 1.254]	0.000
RF invariant	0.9019 (0.0016)	2.664 (0.013)	1.256 [1.242, 1.270]	0.000
variable, $m = 12$, scale_x2				
IN-ARCP	0.9020 (0.0017)	6.691 (0.030)	1.000 [1.000, 1.000]	0.000
AR+raw-RMS	0.9018 (0.0017)	6.699 (0.030)	0.999 [0.996, 1.001]	0.000
AR+global	0.6222 (0.0025)	3.217 (0.016)	2.080 [2.059, 2.101]	0.000
Student	0.8892 (0.0010)	6.422 (0.009)	1.042 [1.033, 1.050]	0.000
Ad-EffOrr native	0.7501 (0.0031)	3.659 (0.028)	1.829 [1.801, 1.856]	0.000
Ad-EffOrr invariant	0.9016 (0.0016)	6.000 (0.031)	1.115 [1.105, 1.126]	0.000
CQR invariant	0.9007 (0.0016)	5.389 (0.028)	1.242 [1.229, 1.254]	0.000
RF invariant	0.9019 (0.0016)	5.328 (0.026)	1.256 [1.242, 1.270]	0.000

7 Every evaluated budget split

A star marks the prespecified approximation-based recommendation. All rows use 1,000 repetitions. Coverage varies slightly with the integer conformal rank; 150/150 has rank coverage 136/151, while the rank-aligned candidates shown have .9. This discrete difference is retained in the planning objective.

κ	N	n	Coverage (MCSE)	Length (MCSE)	vs. equal [95% MC]
1.0	21	279	0.9008 (0.0007)	2.4540 (0.0063)	0.9957 [0.9908, 1.0006]
1.0	51	249	0.9009 (0.0007)	2.4465 (0.0066)	0.9927 [0.9881, 0.9972]
1.0	61*	239	0.9011 (0.0007)	2.4482 (0.0067)	0.9933 [0.9889, 0.9978]
1.0	71	229	0.9012 (0.0007)	2.4498 (0.0068)	0.9940 [0.9896, 0.9984]
1.0	101	199	0.9017 (0.0008)	2.4555 (0.0073)	0.9963 [0.9926, 1.0001]
1.0	150	150	0.9017 (0.0009)	2.4646 (0.0083)	1.0000 [1.0000, 1.0000]
1.0	201	99	0.9016 (0.0010)	2.4809 (0.0102)	1.0066 [1.0015, 1.0117]
1.0	251	49	0.9006 (0.0014)	2.5306 (0.0151)	1.0268 [1.0163, 1.0373]
1.5	21	279	0.8999 (0.0007)	2.3519 (0.0058)	1.0001 [0.9954, 1.0048]
1.5	51	249	0.8998 (0.0007)	2.3392 (0.0060)	0.9947 [0.9904, 0.9990]
1.5	71*	229	0.8999 (0.0007)	2.3385 (0.0061)	0.9944 [0.9903, 0.9985]
1.5	101	199	0.8997 (0.0008)	2.3364 (0.0066)	0.9935 [0.9899, 0.9971]
1.5	150	150	0.9004 (0.0009)	2.3517 (0.0078)	1.0000 [1.0000, 1.0000]
1.5	201	99	0.8988 (0.0011)	2.3545 (0.0099)	1.0012 [0.9960, 1.0064]
1.5	251	49	0.9006 (0.0014)	2.4242 (0.0143)	1.0308 [1.0207, 1.0409]

8 Coefficient MSE diagnostic

The limit is $(1 - \rho^2)\kappa/m$ at $\rho = .8$. Every row uses 10,000 fits. Convergence is not monotone or uniformly accurate at the evaluated finite sample sizes.

m	κ	N	$\widehat{NMSE}$	MCSE	Limit
4	1.0	20	0.09713	0.00156	0.09000
4	1.0	100	0.09319	0.00133	0.09000
4	1.0	500	0.09156	0.00132	0.09000
4	1.5	20	0.12750	0.00199	0.13500
4	1.5	100	0.13053	0.00197	0.13500
4	1.5	500	0.13225	0.00191	0.13500
4	3.0	20	0.16757	0.00284	0.27000
4	3.0	100	0.19780	0.00387	0.27000
4	3.0	500	0.21742	0.00363	0.27000
12	1.0	20	0.03210	0.00050	0.03000
12	1.0	100	0.03079	0.00045	0.03000
12	1.0	500	0.03084	0.00044	0.03000
12	1.5	20	0.04191	0.00064	0.04500
12	1.5	100	0.04468	0.00067	0.04500
12	1.5	500	0.04511	0.00064	0.04500
12	3.0	20	0.05513	0.00097	0.09000
12	3.0	100	0.06510	0.00111	0.09000
12	3.0	500	0.07558	0.00160	0.09000

9 Post-validation optimizer sensitivity

The underlying Gaussian datasets, calibration sets and test episodes are identical to the primary comparison. “Invariant” is one for RMS-invariant preprocessing and zero for the native inputs. “Unscaled” denotes the upstream polynomial expression evaluated with vectorized reductions; it is stored as `canonical` in the CSV. Each cell has 100 repetitions. The main experiment’s simplified/1,000-step rows are retained alongside the three additional settings. These checks address arithmetic and iteration sensitivity for the specified linear construction, not optimal tuning or exact reproduction of the authors’ benchmark.

m	Invariant	Arithmetic	Steps	Coverage (MCSE)	Length	IN/method [95% MC]
4	0	unscaled	1000	0.8968 (0.0020)	2.010	1.153 [1.140, 1.166]
4	0	simplified	1000	0.8975 (0.0020)	2.017	1.149 [1.136, 1.162]
4	0	unscaled	2000	0.8967 (0.0020)	2.009	1.153 [1.140, 1.166]
4	0	simplified	2000	0.8974 (0.0020)	2.017	1.149 [1.136, 1.162]
4	1	unscaled	1000	0.8997 (0.0018)	2.773	0.836 [0.826, 0.846]
4	1	simplified	1000	0.9002 (0.0017)	2.774	0.835 [0.825, 0.845]
4	1	unscaled	2000	0.8996 (0.0018)	2.772	0.836 [0.826, 0.846]
4	1	simplified	2000	0.9001 (0.0017)	2.773	0.836 [0.826, 0.845]
12	0	unscaled	1000	0.8980 (0.0019)	2.049	0.980 [0.971, 0.989]
12	0	simplified	1000	0.8975 (0.0018)	2.044	0.983 [0.974, 0.991]
12	0	unscaled	2000	0.8986 (0.0018)	2.051	0.979 [0.970, 0.988]
12	0	simplified	2000	0.8978 (0.0018)	2.046	0.982 [0.973, 0.990]
12	1	unscaled	1000	0.8989 (0.0016)	2.386	0.842 [0.833, 0.851]
12	1	simplified	1000	0.9001 (0.0015)	2.387	0.841 [0.832, 0.850]
12	1	unscaled	2000	0.8988 (0.0016)	2.385	0.842 [0.833, 0.851]
12	1	simplified	2000	0.9000 (0.0015)	2.386	0.841 [0.833, 0.850]

10 Implementation sequence and algorithm checks

The main manuscript’s Algorithms 1 and 2 specify the mathematical operations and integer planning search. Figure 1 complements them by showing the existing Python call order. Its lifelines denote a caller, an `INARCP` instance, and the module containing `history_scale`; they do not introduce a distributed service or additional data-processing component.

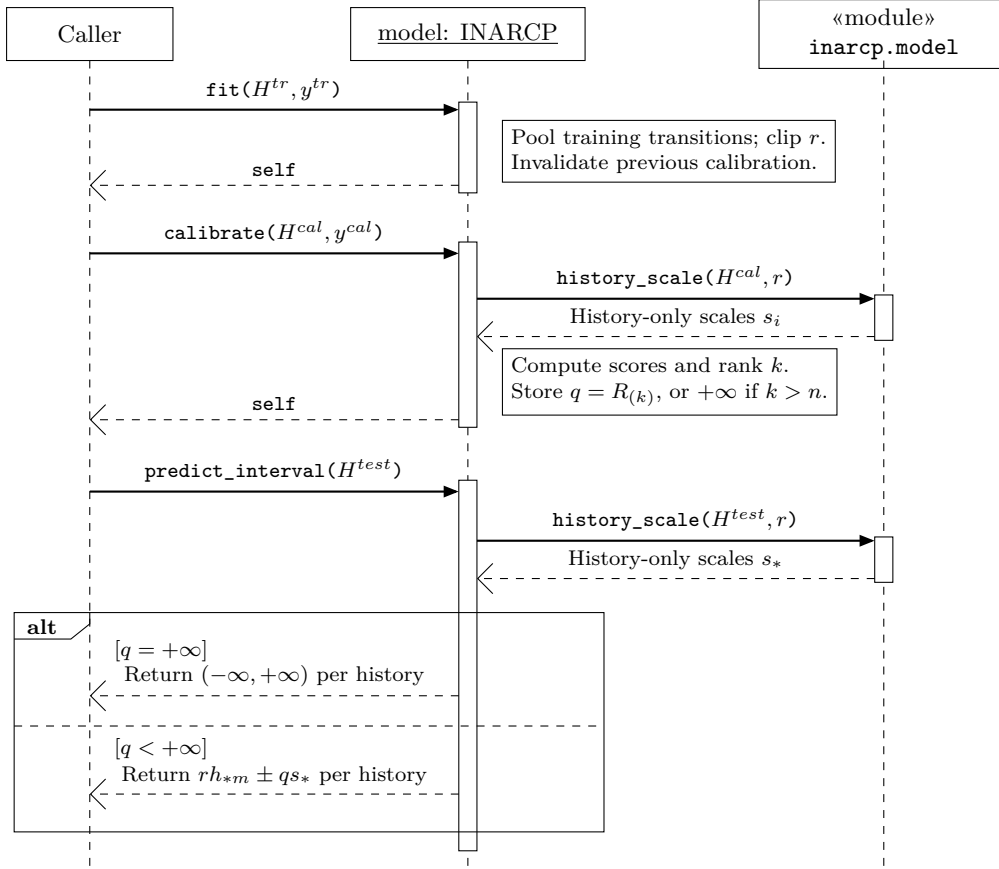


Figure 1: UML sequence diagram for valid inputs. Solid arrows denote synchronous calls; dashed arrows denote returns; activation rectangles mark execution; the **alt** frame selects one prediction return. Training uses observed training responses; calibration uses separate observed calibration responses. Prediction receives test histories only. Both finite and infinite-threshold prediction paths validate the test histories and compute their scales first, as in the code. Input-validation failures terminate the call and are omitted from this success-path diagram. Refitting invalidates calibration and requires a subsequent calibration call before prediction.

The executable check `reproducibility/validate_algorithm_description.py` compares the direct pseudocode formulas with the public implementation, including finite ranks, an unattainable rank, clipping, and the planning search. It also records normalization calls to verify their order, checks rejection of a zero test history even with an infinite threshold, and checks calibration invalidation after refitting. The check is a consistency test of this description, not a new statistical experiment or a proof of independence between episodes.